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PARTIAL EUROPEAN SEARCH REPORT

under Rule 64, paragraph 1 of the European Patent Convention

Application Number EP 22 18 8672

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
X Y	US 2018/348343 A1 (ACHOUR MAHA [US] ET AL) 6 December 2018 (2018-12-06) * paragraphs [0023], [0082], [0077]; figure 9 *	1-3,8,9, 14,15 4,5	INV. G01S7/41 G06N3/04 G06N3/08
Y	CHEIKH K ET AL: "Application of neural networks to radar signal detection in <E1>K</E1>-distributed clutter", IEE PROCEEDINGS: RADAR, SONAR & NAVIGATION, INSTITUTION OF ELECTRICAL ENGINEERS, GB, vol. 153, no. 5, 19 October 2006 (2006-10-19), pages 460-466, XP006027295, ISSN: 1350-2395, DOI: 10.1049/IP-RSN:20050103 * page 464, column 2 - page 465, column 1 *	4,5	ADD. G01S13/89 G01S13/931 G01S13/72 G01S13/34

TECHNICAL FIELDS SEARCHED (IPC)

G01S
G06N

LACK OF UNITY OF INVENTION

The Search Division considers that the present European patent application does not comply with the requirements of unity of invention and relates to several inventions or groups of inventions, namely:

see sheet B

The present partial European search report has been drawn up for those parts of the European patent application which relate to the invention first mentioned in the claims.

Place of search	Date of completion of the search	Examiner
Munich	21 December 2022	Ferrara, Michele

CATEGORY OF CITED DOCUMENTS

X : particularly relevant if taken alone
Y : particularly relevant if combined with another document of the same category
A : technological background
O : non-written disclosure
P : intermediate document

T : theory or principle underlying the invention
E : earlier patent document, but published on, or after the filing date
D : document cited in the application
L : document cited for other reasons

& : member of the same patent family, corresponding document

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LACK OF UNITY OF INVENTION
SHEET B

Application Number
EP 22 18 8672

The Search Division considers that the present European patent application does not comply with the requirements of unity of invention and relates to several inventions or groups of inventions, namely:

1. claims: 1-5, 8, 9, 14, 15

generating the first radar image by masking a range-angle image (RAI) or range-Doppler image (RDI) based on an output of an order statistics (OS) constant false alarm rate (CFAR) (OS-CFAR) detector

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2. claims: 6, 7

generating a heatmap with a tracker of the radar processing chain, and wherein the first radar processing output comprises the heatmap

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3. claims: 10-13

after training the hyperparameter selection neural network using imitation learning, training the hyperparameter selection neural network using reinforcement learning

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The reasons for which the application is deemed to lack unity of invention are given in the attached provisional search opinion. The reader is referred to section [...] of such provisional search opinion.

ANNEX TO THE EUROPEAN SEARCH REPORT ON EUROPEAN PATENT APPLICATION NO. EP 22 18 8672

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report. The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

21-12-2022

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2018348343 A1	06-12-2018	US 2018348343 A1	06-12-2018
		US 2021286050 A1	16-09-2021